The symmetry is broken, once again, by the fact that in our particular world, interactions happen when systems are nearby in space. This is dynamical locality at work. From a Many-Worlds perspective that treats quantum states as fundamental and everything else as emergent, this suggests that we should really turn things around: “positions in space” are the variables in which interactions look local. Space isn’t fundamental; it’s just a way to organize what’s going on in the underlying quantum wave function.

This point of view helps us understand why the Everettian wave function can naturally be divided into a set of approximately classical worlds. This issue is known as the *preferred-basis problem*. Many-Worlds is based on the fact that the wave function of the universe will generally describe all sorts of superpositions, including states where macroscopic objects are in superpositions of being in very different locations. But we never see chairs or bowling balls or planets in superpositions; as far as our experience is concerned, they always seem to have definite locations, and their motion obeys the rules of classical mechanics to a very good approximation. Why don’t the states we see ever involve macroscopic superpositions? We can write the wave function as a combination of many distinct worlds, but why divide it up into *these* worlds in particular?

The answer was essentially figured out in the 1980s, using decoherence, although researchers are still hammering out the details. To get there, it’s useful to turn to that old thought-experiment standby, Schrödinger’s Cat. We have a sealed box containing a cat and a container of sleeping gas. Schrödinger’s original scenario involved poison, but there’s no reason we have to imagine killing the cat. (His daughter Ruth once mused, “I think my father just didn’t like cats.”)

Our experimenter has rigged a spring to pull open the container, releasing the gas and putting the cat to sleep, but only when a detector such as a Geiger counter clicks upon detecting a particle of radiation. Next to the detector is a radioactive source. We know the rate at which particles are emitted from the source, so we can calculate the probability that the counter will click and release the hammer after any given period of time.